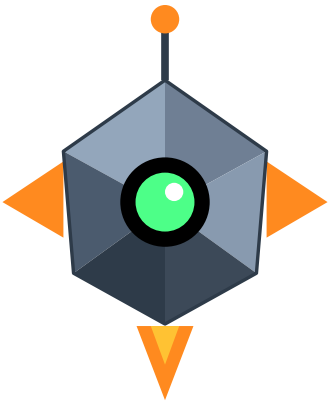
BOOK 2

The Work That Vanishes

The missing corner, and why it works



MULTIX // HELLO
I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

CHAPTER 1

The Missing Corner



MULTIX // CIRCLE CHECK
A circle holds a piece that belongs together, and circles can sit inside circles. A circle is *ready* when everything needed for its calculation is known and its operation is allowed by the current rules. To **pop** a ready circle, replace it with its value and leave everything outside it alone. To **redraw**, move circles around or split a number, as long as the value stays the same. The full rules are in the introduction.

STORY

Ines was making a quilt, and she wanted to know how many squares of cloth she needed. The quilt had 8 rows with 12 squares in each row. Counting every square would take a while, and she did not feel like multiplying 8 by 12 in her head. So she drew it on grid paper instead, and something strange happened.

STEP 1 Cut and slide

Ines drew a rectangle 12 squares long and 8 squares tall. Then she cut off a strip 2 squares wide from the long end. That strip is 2 wide and 8 tall.

She turned the strip on its side, so that it was 8 long and 2 tall, and laid it along the top of what was left.

What she had now was almost a perfect square. The big piece was 10 wide and 8 tall. The strip added a band 2 tall on top, but the band was only 8 long, while the piece underneath was 10 wide. So the band left a small hole in one corner, 2 by 2.

Ines had a 10 by 10 square with a 2 by 2 corner missing.

That is easy to count:

10 times 10 is 100. Take away the missing corner, which is 2 times 2, or 4.

100 minus 4 is 96.

So 8 times 12 is 96. Check it any way you like. It works.

IN CIRCLES

Ines's cut-and-slide can be written with circles. The cutting and sliding is a **redraw**: it changes where the circles are, but not the value.

Try it! Cut a 12 by 8 rectangle out of grid paper. Then do the slide yourself.

TRACE

```
( 8 × 12 )
redraw  →  ( ( 10 × 10 ) − ( 2 × 2 ) )
pop ( 10 × 10 )  →  100
pop ( 2 × 2 )  →  4
( 100 − 4 )
pop  →  96
```

All the thinking happens in the redraw. After that, every circle is easy to pop: the big square is one circle, and the missing corner is another.



MULTIX//SCAN
Provisional answer after the square: 100, with 4 still owed. After the last pop: nothing owed.

STEP 2 Does it work for bigger numbers?

Ines wondered if she had found a one-time accident or a real trick. So she tried a bigger quilt.

18 rows of 22. The middle number between 18 and 22 is 20. Both numbers are 2 away from it. So Ines pictured a 20 by 20 square with a 2 by 2 corner missing.

20 times 20 is 400. Take away 4.

18 times 22 is 396.

48 rows of 52. The middle number between 48 and 52 is 50. Both are 2 away again.

50 times 50 is 2500. Take away 4.

48 times 52 is 2496.

Look at what happened. The numbers got much bigger, but the corner stayed exactly the same size. The part that looked hard, multiplying 48 by 52, turned into a square you can do in your head and a corner you already knew. That is why this chapter is called the work that vanishes: the middle pieces that would have taken so long to add up cancel out before you ever have to count them.

PREDICT FIRST

What would 98 times 102 be? Before you calculate anything, decide how big the corner will be. (It is 2 times 2 again.)

STEP 3 When another route is easier

Now Ines tried 8 rows of 13.

The middle number between 8 and 13 is 10 and a half. That is not a whole number, and her grid paper does not have half squares. Both 8 and 13 are 2 and a half away from the middle, so the corner is no longer a tidy 2 by 2.

So the trick, as she first used it, is not the easiest tool here. She asked what she already knew.

She already knew that 8 times 12 is 96. And 8 rows of 13 is just one more column than 8 rows of 12. One more column is 8 more squares.

96 plus 8 is 104.

So 8 times 13 is 104.

This is a different kind of fix. Instead of finding a perfect square, Ines started from a nearby answer she trusted and added the leftover piece.

FOR BRAVE READERS

if you are happy working with halves, the square trick does still work. 10 and a half times 10 and a half is 110 and a quarter. The corner is 2 and a half times 2 and a half, which is 6 and a quarter. 110 and a quarter minus 6 and a quarter is 104. Same answer, so both roads lead to the same place.

SPLIT · BALANCE · CORRECT

This chapter used the same three moves as the series strategy. We **split** the numbers around a friendly middle, **balanced** them into a square, and **corrected** for the corner that the balancing changed.

WHAT DID WE KEEP?

Write the answers in your notebook.

- When Ines cut and slid the strip, **what stayed the same?** (The total number of squares. We only moved them around.)
- **What did the size of the missing corner depend on?** (Only how far the two numbers were from the middle. It did not matter how big the numbers were.)

WHAT MAKES THIS CORRECTION TRUSTWORTHY?

The square-minus-corner trick is an **exact** rule. Every square in the rectangle is still there, just in a new shape, and the corner counts exactly the squares that are missing. There is no guessing in it.

Every pair of numbers has a middle, so the rule can always be used. What changes is how easy it is. It is most convenient when the middle and the corner are easy to work out. For 8 and 13 the middle is 10 and a half, so adding one more column to 8 × 12 is easier.

PRACTICE

1. Use the trick to find 9 times 11.
2. Use the trick to find 28 times 32.
3. Use the trick to find 98 times 102.
4. Which of these can use the trick with a whole-number middle? For the ones that can, find the answer.
   - 6 times 14
   - 7 times 12
   - 15 times 25
5. Use Ines's other route to find 18 times 23. (Hint: start with 18 times 22, which you already know.)
6. Write **99 × 101** with circles, using 100 as the middle, and pop every circle.

BRIDGE *Bridge from A Useful Wrong Answer:* In the measuring chapter, Mara named one small mistake and counted how many times it happened. Ines does something similar here. She takes an easy answer, a square, names the leftover piece, a corner, and takes it away. In both stories, nothing is guessed at: the correction can be calculated.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 47 × 53 )
2 redraw  →  ( ( 50 − 3 ) × ( 50 + 3 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 3 × 3 ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 3 × 3 )  →  12
6 pop ( 2,500 − 12 )  →  2,488
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 29 × 31 )
2 redraw  →  ( ( 30 − 1 ) × ( 30 + 1 ) )
3 redraw  →  ( ( 30 × 30 ) − ( 1 × 1 ) )
4 pop ( 30 × 30 )  →  900
5 pop ( 1 × 1 )  →  1
6 pop ( 900 − 1 )  →  899
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 98 × 102 )
2 redraw  →  ( ( 100 − 2 ) × ( 100 + 2 ) )
3 redraw  →  ( ( 100 × 100 ) − ( 2 × ( 2 + 1 ) ) )
4 pop ( 100 × 100 )  →  10,000
5 pop ( 2 × ( 2 + 1 ) )  →  6
6 pop ( 10,000 − 6 )  →  9,994
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 2

Putting the Corner Back

STORY

In chapter 1, Ines turned a product into a square with a corner missing. Now she wants to go the other way. She is making a quilt with 35 rows of 35 squares, and she wants to know how many squares that is.
35 × 35 is a square, not a rectangle. Can the missing-corner trick help?

STEP 1 Run the trick backward

In chapter 1, the rectangle 8 × 12 was a 10 by 10 square with a 2 by 2 corner missing. That means:

square = rectangle + corner.

Ines turned that around. A 35 by 35 square is a rectangle plus a corner. Which rectangle? Take 5 off one side and add 5 to the other: 30 × 40. And the corner is 5 × 5.

Here is her notebook:

TRACE

```
( 35 × 35 )
redraw  →  ( ( 30 × 40 ) + ( 5 × 5 ) )
pop ( 30 × 40 )  →  1,200
pop ( 5 × 5 )  →  25
( 1,200 + 25 )
pop  →  1,225
```

So 35 × 35 is **1,225**. The hard part, 35 × 35, became an easy rectangle and a small corner.

STEP 2 Choose the corner you like

Ines can pick how far to step away from 35. She stepped 5 each way because 30 × 40 is easy. Other squares want other steps.

49 × 49. Step 1 each way: 48 × 50 is easy, and the corner is 1 × 1.

TRACE

```
( 49 × 49 )
redraw  →  ( ( 48 × 50 ) + ( 1 × 1 ) )
pop ( 48 × 50 )  →  2,400
```

Test it. Try 65×65. Does the answer end in 25?

```
pop ( 1 × 1 )  →  1
( 2,400 + 1 )
pop  →  2,401
```

98 × 98. Step 2 each way: 96 × 100 is easy, and the corner is 2 × 2. That gives 9,600 + 4 = **9,604**.

The rule is always the same. For any number, step away by some amount *d* in both directions, multiply the two ends, and put back the corner *d* × *d*.



MULTIX//SCAN
A square is a rectangle plus a corner. Check on a small one: $6 \times 6 = 4 \times 8 + 2 \times 2 = 32 + 4 = 36$.

STEP 3 A pattern, and the reason for it

Ines tried the squares of numbers ending in 5, stepping 5 each way:

- 15 × 15: 10 × 20 + 25 = 225
- 25 × 25: 20 × 30 + 25 = 625
- 35 × 35: 30 × 40 + 25 = 1,225
- 45 × 45: 40 × 50 + 25 = 2,025
- 55 × 55: 50 × 60 + 25 = 3,025

She noticed something. Every answer ends in 25. And the front part is a product of two neighbors: 1 × 2, 2 × 3, 3 × 4, 4 × 5, 5 × 6, followed by 25.

That is a pattern from five examples, which is evidence. Here is the reason. 30 × 40 is 100 times 3 × 4, and 40 × 50 is 100 times 4 × 5. The rectangle always has a tens digit times the next one, times 100, and the corner is always 5 × 5 = 25.

STEP 4 Do not forget the corner

Dani tried 35 × 35 and said, “30 times 40 is 1,200, so that is the answer.”

It is not. 1,200 is a useful wrong answer, and the error is exactly the corner that Dani left out. The answer is 25 more than Dani’s.

The corner is the part of the work that does not disappear. The other parts, the long sums, are what the trick saves you from.

SPLIT · BALANCE · CORRECT

This chapter used the same three moves as the series strategy. We **split** the numbers around a friendly middle, **balanced** them into a square, and **corrected** for the corner that the balancing changed.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same when Ines redrew the circles?** (The value of the square. It was 35 × 35 before and after.)
- **What changed?** (The pops. A hard multiplication became two easy ones and an addition.)

WHAT MAKES THIS TRUSTWORTHY?

The rule is exact: a square is always the rectangle plus the corner, because the rectangle plus the corner puts every missing square back. You can test it on a small square you know. Try 6 × 6 with a step of 2: 4 × 8 + 2 × 2 = 32 + 4 = 36. It matches.
The pattern for numbers ending in 5 is a second kind of statement. Five examples made Ines suspect it, and the rectangle-plus-corner reason is what makes her sure.

PRACTICE

1. Find **45 × 45** with a step of 5.
2. Find **99 × 99** with a step of 1.
3. Find **75 × 75** with a step of 5.
4. Find **52 × 52**. Which step would you choose, and why?
5. Dani says "35 × 35 is 30 × 40, which is 1,200." What is missing, and how much is it?

BRIDGE *Bridge to the rest of the series:* In chapter 1 the corner was taken away to turn a rectangle into a square. Here it is put back. In A Useful Wrong Answer, Dani's 1,200 is an estimate with a named error, and the error is the corner.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **1,225**. Leave the others blank. What appears?

1200 + 20	18 × 52 + 17 × 17	1263 − 38	1210 + 4	1201 + 26	1255 − 30	22 × 48 + 13 × 13	1232 + 2
1262 − 37	1206 + 19	1235 − 10	1220 + 5	1264 − 39	1240 − 15	1257 − 32	1259 − 34
1189 + 36	23 × 47 + 12 × 12	1252 − 27	1254 − 29	1264 − 39	25 × 49	1251 − 26	7 × 175
6 × 64 + 29 × 29	1206 + 19	25 × 45 + 10 × 10	30 × 40 + 5 × 5	1234 − 9	1242 − 17	1212 + 13	1250 − 25
1199 + 18	1194 + 31	32 × 38 + 3 × 3	1190 + 35	2 × 68 + 33 × 33	1206 + 19	1262 − 37	1222 + 4
1214 + 21	1217 + 3	37 × 37 − 12 × 12	1211 + 14	22 × 48 + 13 × 13	1252 − 27	1219 + 12	1200 + 17
1217 + 11	1226 + 6	1213 + 13	13 × 57 + 22 × 22	1247 − 22	124 × 124 − 119 × 119	1194 + 24	1205 + 29
1225 + 5	1196 + 23	1209 + 27	1211 + 24	2 × 608	1203 + 14	1218 + 13	1213 + 6



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 3

Same Sum, Different Product

STORY

Ines had a rope that was 24 metres long. She wanted to lay it on the ground in a rectangle for a vegetable garden, with the rope going all the way around the edge.
“Every rectangle I make uses the same rope,” said Ines. “Does every one hold the same amount of garden?”

STEP 1 Try some rectangles

The rope goes around all four sides, so one long side plus one short side uses half the rope. That is 12 metres. Ines listed the rectangles whose two sides add up to 12:

Sides	Garden area
6 by 6	36
5 by 7	35
4 by 8	32
3 by 9	27
2 by 10	20
1 by 11	11

The same rope, but very different gardens. The square holds the most.

STEP 2 The missing-corner trick explains it

Ines noticed that 36 is the biggest, and each other area is 36 minus something:

- $35 = 36 - 1$
- $32 = 36 - 4$
- $27 = 36 - 9$
- $20 = 36 - 16$
- $11 = 36 - 25$

The things taken away are 1, 4, 9, 16, 25. Those are 1×1, 2×2, 3×3, 4×4 and 5×5. The missing corner again!

Here is why, with the 4 by 8 rectangle. It is 2 less than 6 on one side and 2 more than 6 on the other:

Your turn. Try a rope of 20 m. Which rectangle holds the most?

TRACE

```
( 4 × 8 )
redraw  →  ( ( 6 − 2 ) × ( 6 + 2 ) )
redraw  →  ( ( 6 × 6 ) − ( 2 × 2 ) )
pop  →  ( 36 − 4 )
pop  →  32
```

Whenever the two sides add up to 12, you can write them as 6 − d and 6 + d for some d. Stepping away from the square always costs a corner of d × d.



MULTIX//SCAN
Same sum, different product. The square always holds the most. Check: $7 \times 13 = 91$ is less than $10 \times 10 = 100$.

STEP 3 What that means for gardeners

The further you step from a square, the more garden you lose. A step of 1 loses 1 square metre, a step of 2 loses 4, a step of 3 loses 9. The loss grows faster and faster.

This is the reason the square is best. It is also why a long thin garden, even with the same rope, holds so little.

Try it with a different rope. If the two sides must add up to 20, the square has sides 10 and 10, and holds 100. A 7 by 13 garden is 3 steps away from the square, so it loses 3 × 3 = 9, and holds 91. Check: 7 × 13 = 91. ✓

STEP 4 A warning

This works when the sides can be any number. If a fence has to use whole metres and the sides must add up to 11, there is no square. Sides 5 and 6 are each half a step from 5.5. The square 5.5 × 5.5 is 30.25, and 5 × 6 is 30, which is 0.25 less. The corner is smaller than 1, but it is still there: it is 0.5 × 0.5 = 0.25.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What was the same for every rectangle?** (The rope, so the sum of the two sides.)
- **What changed?** (The corner we lose, d × d, so the area.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is the same as in chapter 1: a rectangle is a square with a corner missing. Here we found it by trying six rectangles, which is evidence. The redraw $(6 - d) \times (6 + d) = (6 \times 6) - (d \times d)$ is the reason, and it works for any d.

PRACTICE

1. The two sides add up to 30. What is the area of the square? What is the area of a 12 by 18 rectangle? How far is it from the square?
2. Redraw (7×13) as a square with a corner missing and pop every circle.
3. The sides add up to 16. Which rectangle with whole-number sides holds the most?
4. Ines says a 2 by 10 rectangle holds $36 - 16$. Where does the 16 come from?
5. Without multiplying 9×11 directly, use the sum 20 to say what it is.

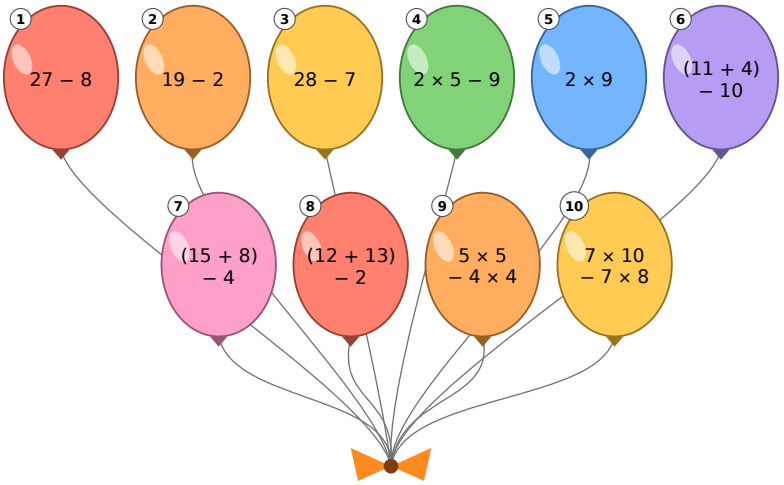
BRIDGE *Bridge to the rest of the series:* In chapter 1 we turned a product into a square. In chapter 2 we turned a square back into a product. Here every product with the same sum is compared with one square. In A Useful Wrong Answer, the square is the easy estimate and $d \times d$ is the named error.

PUZZLE PAGE

The Balloon Message

What did all the gardens agree on? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

										!
1	2	3	4	5	6	7	8	9	10	

CHAPTER 4

Two Corners

STORY

Ines was tiling a floor that was 96 tiles long and 94 tiles wide. The numbers were not close to each other's middle the way 48 and 52 are, but both were close to 100. "Both are a little under 100," said Ines. "Maybe I can use 100 as my friendly number."

STEP 1 Start with the big square

Picture a square 100 tiles by 100 tiles. That is 10,000 tiles. Ines's floor is smaller, 96 by 94. So it is the big square with a strip taken off one side and another strip off the other.

- The strip off the length is 4 wide and 100 long: **4 × 100 = 400** tiles.
- The strip off the width is 6 wide and 100 long: **6 × 100 = 600** tiles.

If Ines takes away both strips, she has removed 10,000 − 400 − 600 = 9,000 tiles' worth.

But wait. The two strips overlap in a small corner, 4 by 6. Ines took that corner away twice. She has to put it back once: 4 × 6 = **24**.

TRACE

```
( 96 × 94 )
redraw  →  ( ( 100 − 4 ) × ( 100 − 6 ) )
redraw  →  ( ( ( 10,000 − 400 ) − 600 ) + ( 4 × 6 ) )
pop  →  ( ( 9,600 − 600 ) + 24 )
pop  →  ( 9,000 + 24 )
pop  →  9,024
```

The floor takes **9,024** tiles.

STEP 2 A shortcut hiding in the middle

Look at 9,000 again. It is 100 times 90, and 90 is 96 − 6. It is also 94 − 4. Ines found that:

- Take one number and subtract the *other* number's shortfall: 96 − 6 = 90 (or 94 − 4 = 90).
- Multiply by 100: 9,000.

Near 100? Count how far under each number is. Those are the shortfalls.

- Add the product of the two shortfalls: 4 × 6 = 24.

TRACE

```
( ( 96 − 6 ) × 100 ) + ( 4 × 6 )   →   9,000 + 24   →   9,024
```

It works because 100 × 100 − 400 − 600 is just 100 × (100 − 4 − 6).



MULTIX // SCAN
Two strips overlap in one corner. Count that corner once. For 96×94 the corner is $4 \times 6 = 24$.

STEP 3 Split, Balance, Correct again

Ines used the series strategy in a new way.

- **Split:** 96 as 100 − 4, and 94 as 100 − 6.
- **Balance:** make a big, friendly square, 100 × 100.
- **Correct:** take off the two strips, then put back the corner that was taken off twice.

Before, the correction was a single corner that was missing. Now there are two strips and a corner that was double-counted.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the redraw keep?** (The value, 96 × 94.)
- **What did we need to correct?** (Two strips, and the corner they share.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is the picture: a big square, two strips, and an overlap that was taken away twice. Every piece of that picture is counted.
The check is another route. Ines found 96 × 94 by taking 96 × 94 = 96 × 100 − 96 × 6 = 9,600 − 576 = 9,024. ✓

PRACTICE

1. Find 97 × 95 by splitting around 100.
2. Find 98 × 93.

3. Ines forgot to put the corner back for 97 × 95 and wrote 9,200. How far off is she, and why?

4. Find 102 × 104. This time both numbers are over 100. The strips are added, not taken off. What happens to the corner?

5. Find 96 × 94 a third way, using the middle trick: 95 × 95 with a corner.

BRIDGE *Bridge to the rest of the series:* In chapter 3, one corner explained the loss from stepping away from a square. Here two strips and a shared corner appear. In A Useful Wrong Answer, the strips are the named errors, and the shared corner is the second-order error that is easy to forget.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  6
6 pop ( 1,600 − 6 )  →  1,594
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × 4 ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × 4 )  →  16
6 pop ( 2,500 − 16 )  →  2,484
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 5

Choosing the Corner

STORY

Ines now knows several tricks. The missing corner, the square with a step, the two strips near 100, and the "one more column" idea from the first chapter. Her next problem is not a new trick. It is choosing the right one.

STEP 1 Look at the numbers first

Before she calculates, Ines looks at the numbers and asks what they are close to.

47 × 53. The numbers are on either side of 50, each 3 away. That is the missing corner.

TRACE

```
( 47 × 53 )
redraw  →  ( ( 50 − 3 ) × ( 50 + 3 ) )
redraw  →  ( ( 50 × 50 ) − ( 3 × 3 ) )
pop  →  ( 2,500 − 9 )
pop  →  2,491
```

97 × 96. Both numbers are a little under 100. That is the two-corner trick: shortfalls 3 and 4.

TRACE

```
( ( 97 − 4 ) × 100 ) + ( 3 × 4 )  →  9,300 + 12  →  9,312
```

18 × 23. The numbers are not close to each other or to 100. But 18 × 22 is something Ines already knows is 396, and 23 is one more column.

TRACE

```
( 18 × 23 )
redraw  →  ( ( 18 × 22 ) + ( 18 × 1 ) )
pop  →  ( 396 + 18 )
pop  →  414
```

13 × 27. Nothing special here. Ines splits one number into friendly pieces.

TRACE

```
( 13 × 27 )
redraw  →  ( ( 10 × 27 ) + ( 3 × 27 ) )
pop  →  ( 270 + 81 )
pop  →  351
```

STEP 2 A little table

Ines made herself a table of what to look for.

Choose first. Four tricks, one product. Pick before you calculate.

If the numbers are…	Use…	Example
on either side of a friendly middle	the missing corner	47 × 53
both just under (or over) 100	two corners	97 × 96
one more or less than a product you know	one more column	18 × 23
none of these	split one number	13 × 27

The tricks do not change the answer. They change how much work it is. All four methods would give the right answer for every product, and Ines picked the cheapest.

STEP 3 Counting the work

How much cheaper? Ines counted pops for 47 × 53.

- Missing corner: 3 pops (50 × 50, 3 × 3, and the subtraction), and the two products are easy.
- Splitting 47 as 40 + 7: 3 pops too (40 × 53, 7 × 53, and the addition), but both products are harder to do in your head.

Both are right. As in The Boundary of a Step, the count depends on the tools you allow, and the number of pops is not the whole story. Here the missing corner has the easier pops.



MULTIX//SCAN
Count the pops and how friendly each one is. Three easy pops can beat three hard ones.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same whichever trick she used?** (The product.)
- **What changed?** (How many pops, and how easy each one was.)

WHAT MAKES THIS TRUSTWORTHY?

Every trick is a redraw that keeps the value, so any of them gives the right product. The trust comes from the reason for each trick, and from checking with a second one. Ines checks 47 × 53 = 2,491 by splitting: 47 × 50 + 47 × 3 = 2,350 + 141 = 2,491. ✓

PRACTICE

Choose a trick for each, say why, and find the product.

1. 38 × 42
2. 99 × 98
3. 24 × 26
4. 19 × 31 (Hint: their sum is 50, so the middle is 25.)
5. Ines picks the “split one number” trick for 47 × 53 and gets 2,491. Is she wrong?

BRIDGE *Bridge to the rest of the series:* In The Choice Before the Calculation, picking the representation comes before calculating. This chapter does that for products. In A Useful Wrong Answer, the second route used as a check is just another trick from this table.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,491**. Leave the others blank. What appears?

2486 + 9	2474 + 29	2486 + 13	2465 + 26	2452 + 39	2473 + 13	2482 + 5	7 × 356
2485 + 2	2474 + 13	2489 + 2	2472 + 19	2499 − 8	2473 + 18	52 × 52 − 15 × 15	2463 + 26
2459 + 26	2505 − 14	2499 − 8	2473 + 18	2510 − 19	53 × 47	2504 − 13	2479 + 23
2518 − 27	2481 + 10	2475 + 16	53 × 47	2494 − 3	2522 − 31	2484 + 7	2459 + 32
2496 − 5	2466 + 25	2500 − 9	2475 + 16	2505 − 14	2523 − 32	2519 − 28	2527 − 36
2517 − 26	2520 − 29	2503 − 12	2486 + 8	2472 + 11	2521 − 30	2467 + 24	2527 − 36
2508 − 17	2500 − 9	2489 + 2	2501 + 2	2470 + 26	2527 − 36	2468 + 23	2466 + 25
2525 − 34	2516 − 25	2505 − 14	2475 + 12	2486 + 14	2453 + 38	2525 − 34	2476 + 15



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 6

Corners with Decimals

STORY

Ines was cutting ribbon. One piece was 4.8 metres long and she needed a rectangle of cloth 4.8 metres by 5.2 metres.
“4.8 times 5.2,” said Ines. “Decimals. Does the missing corner still work?”

STEP 1 The same trick with smaller steps

The two numbers are on either side of 5, each 0.2 away. Ines wrote it the same way she did with whole numbers.

TRACE

```
( 4.8 × 5.2 )
redraw  →  ( ( 5 − 0.2 ) × ( 5 + 0.2 ) )
redraw  →  ( ( 5 × 5 ) − ( 0.2 × 0.2 ) )
pop  →  ( 25 − 0.04 )
pop  →  24.96
```

The cloth is **24.96** square metres.

The missing corner is 0.2 × 0.2. Two tenths of two tenths is four hundredths, so 0.04. It is small, but it is still the corner, and it is still taken away.

STEP 2 Why decimals behave

A decimal like 4.8 is just 48 tenths. Ines noticed that 4.8 × 5.2 is the whole-number product 48 × 52, but in tenths. And she already knows 48 × 52 = 2,496, from chapter 1.

Count places. One place times one place gives two places.

TRACE

```
2,496 hundredths  =  24.96
```

(Tenths times tenths gives hundredths.) The same answer, two ways.

STEP 3 More examples

9.9 × 10.1. On either side of 10, each 0.1 away.

TRACE

```
( ( 10 × 10 ) − ( 0.1 × 0.1 ) )  →  ( 100 − 0.01 )  →  99.99
```

7.5 × 8.5. On either side of 8, each 0.5 away.

TRACE

```
( ( 8 × 8 ) − ( 0.5 × 0.5 ) )  →  ( 64 − 0.25 )  →  63.75
```

0.9 × 1.1. On either side of 1, each 0.1 away.

TRACE

```
( ( 1 × 1 ) − ( 0.1 × 0.1 ) )  →  ( 1 − 0.01 )  →  0.99
```

Look at the last one. 0.9 × 1.1 is a little less than 1. The corner explains exactly how much less: 0.01.

STEP 4 A warning about places

The most common slip is the corner. 0.2 × 0.2 is **0.04**, not 0.4 and not 0.004. A good habit is to count the decimal places: one place times one place gives two places.



MULTIX//WARNING

0.2 × 0.2 is 0.04. Not 0.4. Not 0.004.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same when the numbers got decimals?** (The idea: a rectangle is a square with a corner missing.)
- **What changed?** (The corner got smaller, and the decimal places had to be counted.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is the same as before. The redraw (5 − 0.2) × (5 + 0.2) = (5 × 5) − (0.2 × 0.2) works for any step, whole or decimal.
The check is the whole-number version in tenths: 48 × 52 = 2,496, and 2,496 hundredths is 24.96. ✓

PRACTICE

1. Find 3.8 × 4.2.
2. Find 5.9 × 6.1.
3. Find 1.5 × 2.5. (Hint: the middle is 2 and the step is 0.5.)
4. Find 19.8 × 20.2.
5. Check 3.8 × 4.2 by using whole numbers in tenths.

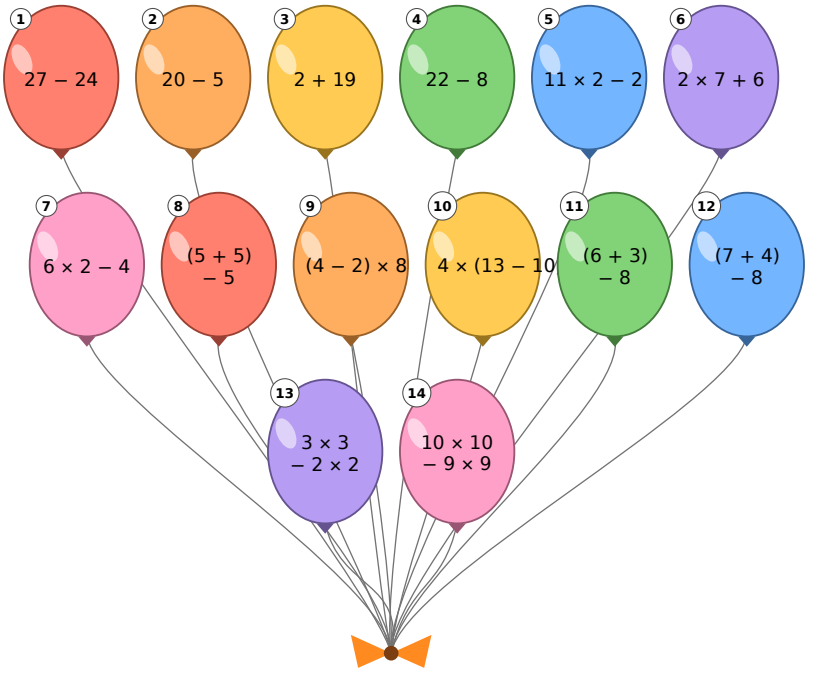
BRIDGE *Bridge to the rest of the series:* In chapter 5, Ines chose the right trick for each product. Here the trick is the same and the numbers change. In A Useful Wrong Answer, 0.04 is a very small correction, and the share question applies: 0.04 out of 24.96 is about 1 in 600.

PUZZLE PAGE

The Balloon Message

What is the first rule for decimals? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

														!
1	2	3	4	5	6	7	8	9	10	11	12	13	14	

CHAPTER 7

Climbing from One Square to the Next

STORY

Ines knew that 25 × 25 is 625. Her teacher asked her for 26 × 26.
“I do not need to start over,” said Ines. “I already have the 25 by 25 square. I just need to build the next one around it.”

STEP 1 Add an L-shaped border

A 26 by 26 square is a 25 by 25 square with an L-shaped border added on two sides.

- One side of the border is 25 long.
- The other side is 25 long.
- The corner is 1 square.

So the border has 25 + 25 + 1 = **51** squares. In circles:

TRACE

```
( 26 × 26 )
redraw  →  ( ( 25 × 25 ) + ( ( 25 + 25 ) + 1 ) )
pop  →  ( 625 + 51 )
pop  →  676
```

26 × 26 is **676**.

Notice the border: 25 + 25 + 1 is also 25 + 26. It is the two squares’ sides added together.

STEP 2 The rule for any square

Next square. From $n \times n$, add $n + (n + 1)$.

To get from one square to the next, add the old side and the new side.

TRACE

```
( 41 × 41 )  =  ( 40 × 40 ) + ( 40 + 41 )  =  1,600 + 81  =  1,681
( 51 × 51 )  =  ( 50 × 50 ) + ( 50 + 51 )  =  2,500 + 101 =  2,601
( 100 × 100 ) =  ( 99 × 99 ) + ( 99 + 100 )  so  ( 99 × 99 ) = 10,000 - 199 = 9,801
```

The last line runs the rule backward. If you know 100 × 100, you can step down to 99 × 99 by taking the border away.

Did 1,681 look familiar? It was the number that broke Kofi's prime recipe in The Order of Discovery. That recipe took n × n, added n, then added 41. At n = 40 it was 1,600 + 40 + 41 = 1,681. That is exactly the border from 40 × 40 to 41 × 41, and it makes 41 × 41. A pattern that looked like a prime recipe was secretly a square.



MULTIX//SCAN
25×25 = 625. The border for 26×26 is 25+26 = 51.
So 26 × 26 = 676.

STEP 3 Why the border is always odd

The border from n × n to the next square is n + (n + 1), which is 2n + 1. That is always an odd number. So the squares go up by the odd numbers, 3, 5, 7, 9, 11 and so on, which is the pattern from Kofi's notebook.

TRACE

```
1,  4,  9,  16,  25,  36
  +3  +5   +7   +9   +11
```

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed when we built the next square?** (The old square, whole, inside the new one.)
- **What did we add?** (An L-shaped border, which is the only new work.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is the picture: a bigger square is the smaller square plus a border, and every square in the border is counted once. The check is to compare with another way, such as the missing corner: 26 × 26 = (25 × 27) + 1 = 675 + 1 = 676. ✓

PRACTICE

1. Find 31 × 31 from 30 × 30.
2. Find 61 × 61 from 60 × 60.
3. Find 20 × 20 from 19 × 19 = 361.

4. Find 49 × 49 from 50 × 50 by stepping down.
5. Check 26 × 26 using the missing corner.

BRIDGE *Bridge to the rest of the series:* In chapter 2, a square was a rectangle plus a corner. Here one square is the next smaller square plus a border. In The Order of Discovery, the border is the reason the odd numbers add up to squares, and it also explains why a prime recipe quietly turned into a square.
Next in this book: Chapter 8: The Cross Pieces That Cancel.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 15 × 17 )
2 redraw  →  ( ( 16 − 1 ) × ( 16 + 1 ) )
3 redraw  →  ( ( 16 × 16 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 16 × 16 )  →  256
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 256 − 2 )  →  254
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 35 × 45 )
2 redraw  →  ( ( 40 − 5 ) × ( 40 + 5 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 5 × 5 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 5 × 5 )  →  30
6 pop ( 1,600 − 30 )  →  1,570
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 58 × 62 )
2 redraw  →  ( ( 60 − 2 ) × ( 60 + 2 ) )
3 redraw  →  ( ( 60 × 60 ) − ( 2 × 2 ) )
4 pop ( 60 × 60 )  →  3,600
5 pop ( 2 × 2 )  →  4
6 pop ( 3,600 − 4 )  →  3,596
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 8

The Cross Pieces That Cancel

STORY

Ines had used the missing-corner trick for **48 × 52** many times. Today she wanted to know *why* the work vanishes. So she took the problem apart, all the way down, and looked at every piece.

STEP 1 Break both numbers

Ines wrote 48 as 50 − 2 and 52 as 50 + 2. Each number now has two parts, and multiplying them means every part of the first meets every part of the second. That gives four small products.

TRACE

```
( ( 50 − 2 ) × ( 50 + 2 ) )
redraw  →  ( ( 50 × 50 ) + ( 50 × 2 ) − ( 2 × 50 ) − ( 2 × 2 ) )
```

This redraw keeps the value, because it only shares out the multiplication across the parts. Now every circle is easy to pop:

TRACE

```
pop ( 50 × 50 )  →  2,500
pop ( 50 × 2 )   →  100
pop ( 2 × 50 )   →  100
pop ( 2 × 2 )    →  4
( 2,500 + 100 − 100 − 4 )
```

STEP 2 Watch the cross pieces

Look at the middle two. One is **+100** and the other is **−100**. They are the same size and opposite, so together they add nothing.

Case file. 48×52 appears in every book.

TRACE

```
( 2,500 + 100 − 100 − 4 )
redraw  →  ( 2,500 − 4 )      the +100 and −100 cancel
pop  →  2,496
```

That is the work that vanishes. The two cross pieces were the hard-looking part of multiplying 48 by 52, and they cancelled each other before Ines ever had to deal with them.



MULTIX // SCAN
The cross pieces cancel only when both numbers are the same distance from the middle.

STEP 3 When the cross pieces do not cancel

Ines wondered if the cancelling always happens. She tried **48 × 53**, where the two numbers are *not* the same distance from 50. 48 is 2 below 50, and 53 is 3 above.

TRACE

```
( ( 50 − 2 ) × ( 50 + 3 ) )
redraw  →  ( ( 50 × 50 ) + ( 50 × 3 ) − ( 2 × 50 ) − ( 2 × 3 ) )
```

The pieces are 2,500, **+150**, **−100** and −6. The cross pieces are 150 and 100. They are different sizes, so they do not cancel:

TRACE

```
( 2,500 + 150 − 100 − 6 )  →  2,544
```

Check it directly: 48 × 53 = 2,544. ✓

So the cancelling is not magic. It happens only when the two numbers are the **same distance** from the middle, one below and one above. Then the two cross pieces are the same size, one added and one taken away.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did breaking the numbers apart keep?** (The value, since the redraw only shared out the multiplication.)
- **What made two pieces cancel?** (The same distance from the middle, one below and one above.)

WHAT MAKES THIS TRUSTWORTHY?

Ines can explain the trick from the four pieces, so it is not a pattern from a few examples. She can also say exactly when it stops working, which is when the distances differ. A rule that tells you where it stops is a rule you can trust.

PRACTICE

1. Break **29 × 31** into four pieces around 30. Which two pieces cancel? What is the answer?
2. Break **47 × 53** into four pieces around 50. Find the answer.
3. Use four pieces to find **98 × 102**.
4. Try **48 × 54** around 50, with distances 2 and 4. Do the cross pieces cancel? Find the answer and check it.
5. Ines says the corner is 2 × 2 because the numbers are 2 away. For 47 × 53, how far away are they? What is the corner?

BRIDGE *Bridge to the rest of the series:* In A Useful Wrong Answer, the corner is the debt on an estimate. In Solving by Changing the Question, sliding two numbers in opposite directions by the same amount keeps their sum and changes the product by exactly the corner.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,496**. Leave the others blank. What appears?

2467 + 25	33 × 76	2463 + 24	2535 − 39	2500 − 4	46 × 54	2476 + 25	2481 + 14
2472 + 17	2462 + 29	2477 + 27	2475 + 21	2471 + 25	2472 + 23	3 × 834	2478 + 24
2472 + 12	2462 + 34	2472 + 24	2458 + 38	2522 − 26	2505 − 9	2461 + 35	2488 + 2
2489 + 7	2535 − 39	2481 + 27	2475 + 21	2531 − 35	2502 + 5	2529 − 33	2533 − 37
2 × 1248	61 × 61 − 35 × 35	52 × 48	2523 − 27	2508 − 12	156 × 16	2492 + 4	52 × 48
2490 + 6	2534 − 38	2478 + 18	2473 + 23	2516 − 20	64 × 64 − 40 × 40	2513 − 17	2475 + 21
15 × 167	24 × 104	2512 − 16	83 × 30	2490 + 16	2462 + 34	2502 − 6	2471 + 24
2479 + 23	2478 + 18	39 × 64	2483 + 5	5 × 499	2531 − 35	110 × 110 − 98 × 98	2480 + 12



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

Answers (turn the book around)

Chapter 1

1. The middle is 10, and the corner is 1 times 1. 100 minus 1 is **99**.
2. The middle is 30, and the corner is 2 times 2. 900 minus 4 is **896**.
3. The middle is 100, and the corner is 2 times 2. 10,000 minus 4 is **9,996**.
4. 6 times 14: middle is 10, corner is 4 times 4 = 16, so 100 minus 16 is **84**. 7 times 12: the middle would be 9 and a half, so the whole-number trick does not apply. 15 times 25: middle is 20, corner is 5 times 5 = 25, so 400 minus 25 is **375**.
5. 18 times 22 is 396, and one more column of 18 gives 396 plus 18, which is **414**.
6. (99 × 101) redraws as ((100 × 100) − (1 × 1)). Pop the two circles to get 10,000 and 1, and then (10,000 − 1) pops to **9,999**.

Spot the Slip.
Notebook A: slip on line **5**. 3 × 3 is 9. The answer should be 2,491.
Notebook B: **perfect**. 29 × 31 = 899.
Notebook C: slip on line **3**. The corner is 2 × 2, not 2 × (2 + 1). The answer should be 9,996.

Chapter 2

1. ((40 × 50) + (5 × 5)) pops to 2,000 + 25 = **2,025**.
2. ((98 × 100) + (1 × 1)) pops to 9,800 + 1 = **9,801**.
3. ((70 × 80) + (5 × 5)) pops to 5,600 + 25 = **5,625**.
4. A step of 2 gives ((50 × 54) + (2 × 2)) = 2,700 + 4 = **2,704**. A step of 2 makes one end 50, which is easy to multiply by.
5. The corner is missing. It is 5 × 5 = **25**, so the answer should be 1,225.

Color by Answer (1,225). The shaded boxes make a heart.



Answers (turn the book around)

Chapter 3

1. The square is 15 by 15, area **225**. A 12 by 18 is 3 steps from 15, so it loses 3 × 3 = 9, giving 216. Check: 12 × 18 = 216. ✓
2. ((10 − 3) × (10 + 3)) → ((10 × 10) − (3 × 3)) → (100 − 9) → **91**.
3. The square, **8 by 8**, which holds 64.
4. 2 is 4 steps below 6 and 10 is 4 steps above. The corner is 4 × 4 = **16**.
5. The sum 20 gives a square of 10 by 10, which is 100. The corner is 1 × 1 = 1, so 9 × 11 is **99**.

Balloon Message.

1.	27 − 8	= 19	S
2.	19 − 2	= 17	Q
3.	28 − 7	= 21	U
4.	2 × 5 − 9	= 1	A
5.	2 × 9	= 18	R
6.	(11 + 4) − 10	= 5	E
7.	(15 + 8) − 4	= 19	S
8.	(12 + 13) − 2	= 23	W
9.	5 × 5 − 4 × 4	= 9	I
10.	7 × 10 − 7 × 8	= 14	N

Message: **SQUARES WIN!**

Chapter 4

1. Shortfalls are 3 and 5. ((97 − 5) × 100) + (3 × 5) = 9,200 + 15 = **9,215**.
2. Shortfalls are 2 and 7. ((98 − 7) × 100) + (2 × 7) = 9,100 + 14 = **9,114**.
3. She is off by **15**, which is the corner 3 × 5 that she forgot to put back.
4. Overs are 2 and 4. ((102 + 4) × 100) + (2 × 4) = 10,600 + 8 = **10,608**. Here the corner is added because it is a new piece outside the big square.
5. 96 × 94 is 95 × 95 with a 1 by 1 corner missing: 9,025 − 1 = **9,024**. ✓

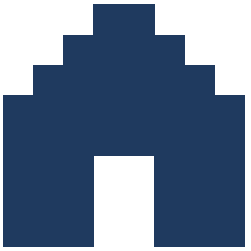
Spot the Slip.
Notebook A: slip on line **5**. 2 × 2 is 4. The answer should be 1,596.
Notebook B: slip on line **3**. The corner is 1 × 1, not 1 × (1 + 1). The answer should be 624.
Notebook C: **perfect**. 46 × 54 = 2,484.

Answers (turn the book around)

Chapter 5

1. On either side of 40 (2 away): (40 × 40) − (2 × 2) = 1,600 − 4 = **1,596**.
2. Both just under 100, shortfalls 1 and 2: ((99 − 2) × 100) + (1 × 2) = 9,700 + 2 = **9,702**.
3. On either side of 25 (1 away): 625 − 1 = **624**.
4. On either side of 25 (6 away): 625 − 36 = **589**.
5. No. All the tricks keep the value, so she gets the right answer. She just did more work than she needed.

Color by Answer (2,491). The shaded boxes make a house.



Chapter 6

1. (4 × 4) − (0.2 × 0.2) = 16 − 0.04 = **15.96**.
2. (6 × 6) − (0.1 × 0.1) = 36 − 0.01 = **35.99**.
3. (2 × 2) − (0.5 × 0.5) = 4 − 0.25 = **3.75**.
4. (20 × 20) − (0.2 × 0.2) = 400 − 0.04 = **399.96**.
5. 3.8 is 38 tenths and 4.2 is 42 tenths. 38 × 42 = 1,596, and 1,596 hundredths is 15.96. ✓

Balloon Message.

1.	27 − 24	= 3	C
2.	20 − 5	= 15	O
3.	2 + 19	= 21	U
4.	22 − 8	= 14	N
5.	11 × 2 − 2	= 20	T
6.	2 × 7 + 6	= 20	T
7.	6 × 2 − 4	= 8	H
8.	(5 + 5) − 5	= 5	E
9.	(4 − 2) × 8	= 16	P
10.	4 × (13 − 10)	= 12	L
11.	(6 + 3) − 8	= 1	A
12.	(7 + 4) − 8	= 3	C
13.	3 × 3 − 2 × 2	= 5	E
14.	10 × 10 − 9 × 9	= 19	S

Message: **COUNT THE PLACES!**

Answers (turn the book around)

Chapter 7

1. 900 + (30 + 31) = 900 + 61 = **961**.
2. 3,600 + (60 + 61) = 3,600 + 121 = **3,721**.
3. 361 + (19 + 20) = 361 + 39 = **400**.
4. 2,500 − (49 + 50) = 2,500 − 99 = **2,401**.
5. (25 × 27) is 675, and adding the corner 1 × 1 gives **676**. ✓

Spot the Slip.
Notebook A: slip on line **3**. The corner is 1 × 1, not 1 × (1 + 1). The answer should be 255.
Notebook B: slip on line **5**. 5 × 5 is 25. The answer should be 1,575.
Notebook C: **perfect**. 58 × 62 = 3,596.

Chapter 8

1. (30 × 30) + (30 × 1) − (1 × 30) − (1 × 1) = 900 + 30 − 30 − 1. The two 30s cancel. The answer is **899**.
2. 2,500 + 150 − 150 − 9. The cross pieces cancel. The answer is 2,500 − 9 = **2,491**.
3. 10,000 + 200 − 200 − 4, so **9,996**.
4. The pieces are 2,500, +200, −100 and −8. They do not cancel. 2,500 + 200 − 100 − 8 = **2,592**. Check: 48 × 54 = 2,592. ✓
5. They are 3 away, so the corner is 3 × 3 = **9**.

Color by Answer (2,496). The shaded boxes make a robot.

